

SKILLS

Languages	Java, C, C++, C#, Python, JavaScript
Frameworks & Libraries	FastAPI, .NET MAUI, scikit-learn, librosa, Ollama, RAG (Retrieval-Augmented Generation), Node.js, REST APIs
Software & Tools	Git, SQLite, pytest, Linux, Claude Code, VS Code, Office Suite
Concepts	Object-Oriented Programming, Data Structures & Algorithms, Machine Learning, Embedded Systems, MVVM Architecture, Version Control, Cross-Platform Development

EDUCATION

Bachelor of Applied Science in Computer Engineering, University of British Columbia	2025 — Present
Bachelor of Science in Chemistry for the Medical Sciences, University of Victoria	2014 — 2018
Engineering First-Year Certificate, Kwantlen Polytechnic University	2024 — 2025
CGPA: 94%	

EXPERIENCE

Embedded Systems & IoT Engineering Co-op	Sept 2026 — Present
UBC Department of Electrical and Computer Engineering, Canadian Coast Guard Project	Vancouver, British Columbia
- Designed and implemented a canonical data model and SQLite persistence layer in Python for an edge-computing gateway service, including measurement normalization and structured validation, covered by 20 passing automated unit and integration tests - Built a local FastAPI dashboard and read-only REST API serving live sensor data with no cloud dependency, including freshness checks and upload-state tracking, validated by 45 passing automated tests - Set up and configured the embedded Linux environment used to run and test the gateway software, including networking and monitoring tooling to support development	

PROJECTS

Phonocardiogram Heart Sound Classifier — Python, scikit-learn, librosa, Claude Code	2026
- Built an end-to-end machine learning pipeline to classify heart sound recordings as Normal/Abnormal from the PhysioNet/CinC 2016 Challenge dataset (3,240 recordings), using Claude Code to accelerate implementation while directing architecture, feature engineering, and validation decisions - Diagnosed and fixed a data leakage bug where peak-detection-based segmentation introduced label-correlated artifacts, replacing it with fixed-length windowing and rigorous train/test isolation to ensure model generalizability - Engineered MFCC audio features with librosa and trained an SVM classifier achieving 0.87 balanced accuracy at the patient level, applying patient-grouped cross-validation to prevent data leakage and reduce overfitting	
Memory Chatbot — Python, Ollama, RAG	2026
- Built a locally-run, privacy-preserving AI chatbot in Python using Ollama with Llama 3, featuring persistent conversation memory serialized to JSON and indexed for retrieval by date - Implemented a Retrieval-Augmented Generation (RAG) pipeline with semantic search and vector similarity to surface relevant context from past conversations and inject it into new LLM queries - Designed a modular command-line interface (CLI) supporting conversation recall, semantic search, and on-demand user profiling via automated summarization	
IoT Network Watchdog — C++, ESP32, JavaScript	2026
- Designed and built an asynchronous, full-stack web dashboard using JavaScript (ES6, Fetch API) and HTML5, serving live network metrics through a lightweight REST-style API with non-blocking backend architecture - Implemented persistent data logging and a lightweight local database layer using the LittleFS filesystem, handling structured event records with NTP-synchronized timestamps - Engineered the backend network-monitoring logic in C++ on an ESP32 microcontroller, tracking connectivity status and latency with minimal system overhead	
UBC Course Planner — C#, .NET MAUI	2025
- Developed a cross-platform, responsive GUI application for course planning using C# and .NET MAUI, implementing the MVVM (Model-View-ViewModel) architecture for maintainable, testable UI code - Implemented data-driven schedule persistence and course conflict-detection logic to improve reliability and user experience	